

EDUCATION

- Sichuan University** Chengdu, China
Bachelor of Biological Sciences 09/2016 - 07/2020
- University of Chinese Academy of Sciences** Beijing, China
Master of Regenerative Medicine 09/2020 - 12/2023
- University of Pittsburgh** Pittsburgh, PA
Ph.D. Student in Human Genetics 08/2024 - Present
Relevant Coursework: Genetics, Molecular Biology, Cell Biology, Calculus, Bioinformatics, Biostatistics, Statistical Genomics, Data Analysis for High Throughput Sequencing, Deep Learning for Natural Language Processing

SKILLS SUMMARY

- Computer skills:** Proficient in the use of shell scripts, R language, and Python-related software
- Bioinformatics:** Proficient in single-cell omics sequencing data analysis process
- Experiment:** Able to independently perform molecular biological experiments, such as western blot, RT-PCR, plasmid construction, etc
- Interpersonal:** Leadership, Communication, Problem-solving, Time Management

WORK EXPERIENCE

- Changping Laboratory** Beijing, China
Bioinformatics Engineer, Xuetao Cao Lab 01/2024 - 07/2024
 - Mainly responsible for single-cell multiome data analysis.

RESEARCH EXPERIENCE

- Project1: Angiogenesis of Colorectal Cancer Stem Cells in Zebrafish** Chengdu, China
Scientific Research Training 02/2017 - 08/2018
 - As the first person in charge completed the innovation and entrepreneurship training project.
 - Master the acquisition and culture of colorectal cancer stem cells; Breeding and microinjection of zebrafish; To observe the growth and migration of colorectal cancer stem cells in early zebrafish embryos.
- Project2: Trackless Somatic Cell Reprogramming** Chengdu, China
Scientific Research Training 09/2018 - 06/2020
 - Completed the construction, transformation, and expression of mCherry and EGFP plasmid vectors, tested the appropriate conditions for protein purification, which were used for the next step of antibody preparation (completed by the company), and tested the titer of the antibody.
- Project3: The Primate-specific Pro-senescent Microglia** Beijing, China
Laboratory Research Topic 09/2020 - 12/2021
 - Participated in the analysis and verification of the single-cell atlas of a primate organ, and used Imaris to construct a 3D model to verify the spatial localization of microglia and CHIT1.
- Project4: Aging Induces Region-specific Dysregulation of Hormone** Beijing, China
- Synthesis in the Primate Adrenal Gland**
Laboratory Research Topic 06/2021 - 03/2024
 - In this project, we aimed to investigate the molecular and cellular changes of primate adrenal aging using single-nucleus transcriptome analysis. Conducted pseudotime analysis of the regeneration process of the adrenal cortex and revealed the molecular mechanism of the reticular zona atrophy during aging.
 - We identified lipid metabolism and steroidogenic pathways as core processes impacted by aging and that centripetal adrenocortical differentiation was dysregulated in aged adrenal tissues. We found that cells in the zona reticularis region, which produces dehydroepiandrosterone sulfate (DHEA-S), were highly susceptible to aging, as reflected by transcriptional disorder, depletion, and a deeply perturbed hormone production. Remarkably, the cholesterol uptake receptor LDLR was downregulated in all cell types of the outer cortex.

- **Project5: Exercise-induced Geroprotective Effects across the Body** Beijing, China
Laboratory Research Topic 10/2021 - 01/2023
 - Single-cell and single-nucleus sequencing of different organs of young and old mice after exercise for one year, responsible for this project in the single-nucleus sequencing part, R and Python are used to complete the preliminary quality control, integration, and subsequent differential analysis, and Seurat is mainly used in combination with other software to analyze its cell type composition, differentially expressed genes, main enrichment pathways, transcriptional noise, cell communication, pseudotime and transcription factors prediction to establish downstream intervention targets.
 - The study found that long-term exercise can significantly reduce the pan-tissue aging degree of old individuals, and enhance the body's anti-infection ability. At the mechanistic level, the up-regulation of the core circadian transcription factor BMAL1 mediated the protective effect of exercise on the central vascular aging process.
- **Project6: Single-nucleus Transcriptome Analysis of Human Synovial Aging** Beijing, China
Laboratory Research Topic 10/2021 - 11/2023
 - In this project, we aimed to investigate the molecular and cellular changes of human synovial aging using single-nucleus transcriptome analysis.
 - Identified VCAM1, a key factor mediating the aggregation of immune cells during synovial aging, and determined its high expression in aged sub-lining fibroblasts. In addition, the promotion of synovial senescence by the decline of the core transcriptional regulator FOXO1 was also found and its effect was verified in vitro models.
- **Project7: Multi-omics Profiling of Peripheral Immune Cells in Alzheimer's** Pittsburgh, PA
Laboratory Research Topic 08/2025 - Present
 - Ongoing Ph.D. research project.

PUBLICATIONS

- Sun S[#], Ma S[#], Cai Y[#], Wang S[#], Ren J[#], Yang Y[#], Ping J, **Wang X**, Zhang Y, Yan H, Li W, Esteban CR, Yu Y, Liu F, Belmonte JCI, Zhang W*, Qu J*, Liu GH*. A single-cell transcriptomic atlas of exercise-induced anti-inflammatory and geroprotective effects across the body. *The Innovation*, 2023, 4(1): 100380.
- Sun S[#], Li J[#], Wang S[#], Li J[#], Ren J[#], Bao Z[#], Sun L[#], Ma X[#], Zheng F, Ma S, Sun L, Wang M, Yu Y, Ma M, Wang Q, Chen Z, Ma H, **Wang X**, Wu Z, Zhang H, Yan K, Yang Y, Zhang Y, Zhang S, Lei J, Teng ZQ, Liu CM, Bai G, Wang YJ, Li J, Wang X, Zhao G, Jiang T, Belmonte JCI, Qu J*, Zhang W*, Liu GH*. CHIT1-positive microglia drive motor neuron ageing in the primate spinal cord. *Nature*, 2023, 624(7992): 611–620.
- Liu F[#], Lu Y[#], **Wang X**[#], Sun S[#], Pan H, Wang M, Wang Z, Zhang W, Ma S, Sun G, Chu Q, Wang S*, Qu J*, Liu GH*. Identification of FOXO1 as a geroprotector in human synovium through single-nucleus transcriptomic profiling. *Protein & Cell*, 2024, 15(6): 441–459.
- Wang Q[#], **Wang X**[#], Liu B[#], Ma S[#], Zhang F[#], Sun S, Jing Y, Fan Y, Ding Y, Xiong M, Li J, Zhai Q, Zheng Y, Liu C, Xu G, Yang J, Wang S, Ye J, Belmonte JCI, Qu J*, Liu GH*, Zhang W*. Aging induces region-specific dysregulation of hormone synthesis in the primate adrenal gland. *Nature Aging*, 2024, 4(3): 396–413.
- Lu Y, **Wang X**, Saibro-Girardi C, Fitz NF, Koldamova R, Lefterov I. RXR-mediated remodeling of transcriptional and chromatin landscapes in APP mouse brain: insights from integrated single-cell RNA and ATAC profiling. *Cells*, 2025, 14(24): 1970.

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HONORS AND AWARDS

- Second prize of the Hebei Mathematics League Fall, 2015
- Second-class individual Scholarship at Sichuan University Spring, 2017
- Excellent Student cadre of Sichuan University Fall, 2017
- First-class individual Scholarship of Sichuan University Spring, 2018
- Excellent President of academic society of Sichuan University Spring, 2019
- Second-class individual Scholarship at Sichuan University Spring, 2019
- University Academic Scholarship of Chinese Academy of Sciences Spring, 2021
- Organized and participated in old book fundraising and public welfare lectures many times, and won the honor of outstanding volunteers